

AHMED OTHMAN QADIR

Communication Engineer



 Twimalik , Sulaimani, Kurdistan Region, Iraq

 ahmad.tce2223096@spu.edu.iq

 +964 770 090 7212

 26/1/2004

Profile

Communication Engineering graduate from Sulaimani Polytechnic University, consistently ranked among the top students — achieving 1st place in my third year. I have hands-on experience with Huawei telecommunications systems through the Asiacell ASAS Internship Program, along with practical exposure to Ericsson equipment, Cisco cybersecurity, and Arduino-based embedded systems. I attended the DTI Summer School at FH Münster, Germany, earning 4.5 ECTS credits. A field visit to Korek Telecom gave me practical exposure to real telecom infrastructure including RBS systems and RF antenna networks. My final-year project integrates AI with CST software to develop a Metamaterial Absorber for cancer and nitrate detection. I am passionate about RF engineering, intelligent systems, and building AI-driven solutions for real-world problems.

Certificate

▪ Design Technology Innovation (DTI) Summer School, FH Münster University of Applied Sciences	2025 Aug – 2025 Sep
	Münster, Germany
- Successfully completed an interdisciplinary summer school featuring lectures, workshops, and practical exercises. - Covered robotics, AI, VR, coding, 3D printing, prototyping, sustainability, and business innovation. - Earned 4.5 ECTS credit points upon successful completion.	
▪ Asas Internship Program , Asiacell ASAS	2025 Jul – 2025 Aug
- I gained practical familiarity with telecommunications components and systems used by Asiacell, including several Huawei technologies. - I observed and helped address multiple technical issues encountered during the training period.	
▪ ICT and Cybersecurity Training Course , SPU CISCO LAB	2024 Aug
I received training in cybersecurity protection, ethical hacking, and data security from instructors from the Polytechnic, Cisco in the United States, Iq security team, and a Ukrainian cybersecurity company through a program organized by Cisco.	
▪ Telecommunication Course, Asoy Gash Institute	2024 Jul – 2024 Aug
- I completed a certified telecommunications course, where I improved my understanding of telecom systems and their practical use. - During the course, I became familiar with the Ericsson system and saw the equipment and troubleshooting methods used to solve technical issues for companies, especially Halabja Group.	
▪ AI Seminars & Scientific Panel, MegaMind Academy & Univ. of Garmian	2026 Mar – 2026 May
Earned 4 certificates across specialized webinars and a panel exploring Artificial Intelligence integration in academic research, learning, and digital security.	

Education

Bachelor of Science in Communication Engineering, Sulaimani Polytechnic University (SPU)

2022 Sep – 2026 Jun

- Ranked **third** in my department in the first year.
- Ranked **second** in my department in the second year.
- Ranked **first** in my department in the third year.
- Ranked **second** in my department in the fourth year.

Project

Smart Anti-Drone RF Defense:

Designed and simulated from scratch to protect sensitive locations in Sulaymaniyah — modeled over Sulaymaniyah International Airport. The system uses four directional Yagi antennas covering 1,600 m², neutralizing drones by transmitting a high-power noise signal at the drone's frequency to instantly sever its control link. An AI layer assesses threat levels, identifies drone types, and monitors proximity in real time — targeting 2.4 GHz and 5.1 GHz while avoiding interference with civilian infrastructure.

Metamaterial Absorber Design for Sensing Applications Enhanced with AI Techniques:

- The Metamaterial Absorber project is **my final-year university project** and one of my most significant works — built entirely from scratch in a way that had not been done before (0 to 100, fully self-developed).

I began by generating a very large dataset of approximately 400,000 data points using CST simulation software for the absorber. I then trained an AI system in Python on this dataset to simplify and accelerate the work of engineers.

The system includes several highly important features that benefit both communication engineers and CST software users:

- Cancer detection — performed directly using the patient's own blood
- Nitrate detection in fruits and food products
- A dedicated AI chatbot tab for answering any questions about the system
- Automated report generation based on the results

Nano-Herbal Formulation Intelligence Platform (NanoHerbalAI)

An AI-powered veterinary decision system built from scratch on 35 peer-reviewed papers, covering antiviral, antimicrobial, antioxidant, anti-inflammatory, and expectorant herbal compounds. The platform features an AI prediction model that estimates efficacy when compounds are combined, an evidence-grounded AI chatbot with full paper-backed traceability, and a references module linking every output to its source. Core modules include Drug Selector, Formulation Builder, Formula Engine, Drug Comparison, Biomarker Analysis, and Stability Timeline.

Arduino & Raspberry Pi Embedded Systems Projects:

Built several hands-on embedded systems projects using Arduino Nano, Uno, and Raspberry Pi programmed in C and Python, including a small radar system, a mobile-controlled remote car, a sensor-driven autonomous car, a digital timer, and a portable public-space alarm system — demonstrating practical experience in microcontroller programming, sensor integration, and real-time hardware control.

Op-Amp Signal Processing Projects:

Built several analog circuit projects using the LM741 operational amplifier IC, including an ECG signal conditioning circuit designed to filter and reduce noise from heart signal recordings, and a function generator capable of producing sine, square, and triangular waveforms — demonstrating solid understanding of analog electronics and signal processing fundamentals.

Skills

- **Telecommunications:** RF systems, antenna engineering, frequency planning, 2G/3G/4G networks
- **Software:** CST Studio, MATLAB, Multisim, Cisco Packet Tracer, AutoCAD, PyCharm, Antigravity IDE
- **AI & Data:** AI system development, data generation & training, prompt engineering
- **Programming:** Python, C (Arduino)
- **Hardware:** Arduino, Raspberry Pi, circuit design & implementation